

Multivariate statistics

Lecture 08:

Principal Components Analysis, PCA

Literature and Problems: See course schedule

Agenda (preliminary):

Motivation for PCA

- dimension reduction
- decorrelation of variables

Problem formulation for $PC_1, PC_2, \dots, PC_p$

Solution: Eigenvalue decomposition

- SLIDE: EVD (and SVD)
- PC_i loadings, variances and covariances
- total variance and variance explained by PC_i

Geometrical interpretation (linear change of basis)

Sample-based PCA on S procedure

- X, S , EVD
- Pareto/scree-plot and truncation
- PC_i loadings, variances and scores
- bi-plot of scores and loadings
- correlation between PC_i and original variable X_k
- applications: simplification, interpretation, pre-analysis for LR/MANOVA
- DEMO: PCA(S) for stock exchange example

Potential pitfalls of PCA on S

Standardization of variables

Sample-based PCA on R procedure

- Z, R , EVD (note: different eigen-values/vectors than for S)
- analogue to PCA on S
- DEMO: PCA(R) for stock exchange example

Large sample approximate distributions of eigen-values/vectors --> inference

Outlier detection by PC's and MVN modelcheck for inferences

SVD(X) based calculation of EVD(S) --> Problem 8.3